Continuous Data Distributions

↓
uncountable values
↳ we go for a particular range (10-50, 1000-2000, etc.)

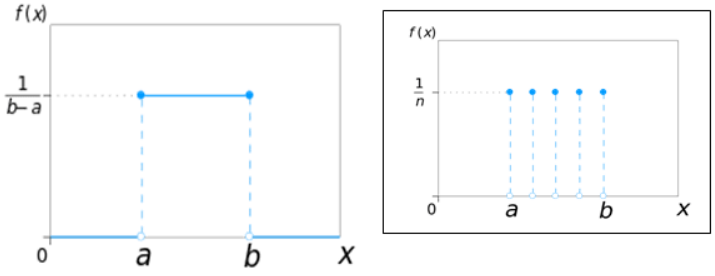
Types of Continuous Distributions :-

(1) Uniform Distribution :-

All outcomes within a specified range are equally likely.

Two types of Uniform Dist :-

a) Discrete Uniform Dist. ⇝ for finite set of values
b) Continuous " " ⇝ for all values within a continous interval (range)



(a) Continuous Uniform Distribution (b) Discrete Uniform Distribution

Characteristics :-

- Range → Distance defined by a lower bound & upper bound
- Probability Density function :- Distribution that provides probability density of a continuous random variable within a specific range of values.

Ex- In a city there a 10 lakh people.
How many people will have their height between 168-172 cm?

Formula :-

(1) for discrete uniform :-

Formula :

① For discrete uniform :-

$$P(X=x) = \frac{1}{n}, \quad \text{for } x \in x_1, x_2, x_3 \dots x_n$$

where,
n = total no. of discrete outcomes

② For continuous uniform :-

$$f(x) = \frac{1}{b-a}, \quad \text{for } x \in [a,b]$$

③ For CDF :- [for continuous]

$$F(x) = \frac{x-a}{b-a}, \quad \text{for } x \in [a,b]$$

- Mean (μ) $\rightsquigarrow$ $\mu = \frac{a+b}{2}$
- Variance (σ^2) $\rightsquigarrow$ $\sigma^2 = \frac{(b-a)^2}{12}$
- Shape $\rightsquigarrow$ It is flat across the interval

Ques. A bus can arrive at a bus stop anytime in b/w 0 mins to 10 mins. What is the probability that it will arrive within 4 mins.

Sol :- Given :- $a = 0$, $b = 10$, $x = 4$

∵ we want the prob. within 4 mins

∴ we calculate CDF

$$f(x) = \frac{x-a}{b-a}$$

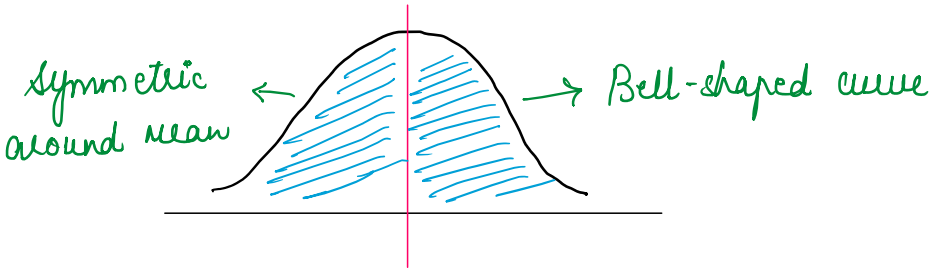
$$F(4) = \frac{4-0}{10-0} = \frac{4}{10}$$

$$F(4) = 0.4$$

Ans:- 40% chances that bus will arrive within 4 mins.

(2) Normal Distribution :-

- It is also known as gaussian distribution.
- It is the most widely used distribution.
- It looks like a bell-shaped curve.
- It is a continuous distribution that is symmetric around the mean.
- Majority of natural phenomenas follows a normal dist-
- Majority of the values are centered around mean and fewer points are found at tails.



Normal Distribution

Formula:-

PDF :-

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\left[\frac{-(x-\mu)^2}{\sigma^2}\right]}$$

where,

μ = mean

σ^2 = variance

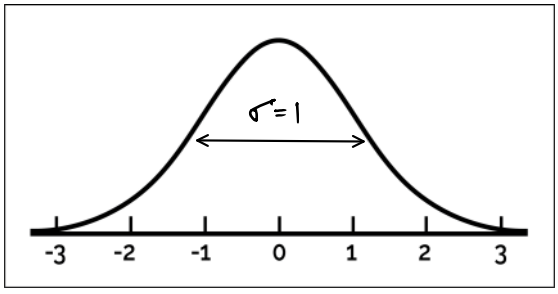
Characteristics:-

- Shape → Symmetric around mean

Standard Normal Distribution :-

A special case of Normal Dist. when,

$\mu = 0$, $\sigma = 1$

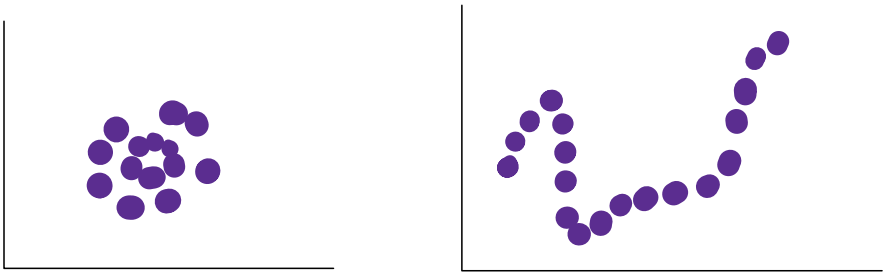


③ Log-Normal Distribution :-

Why we use log? ⇝ A scale that is used to handle extreme values

- very big
- very small

Majority of the data in real word is not normal.



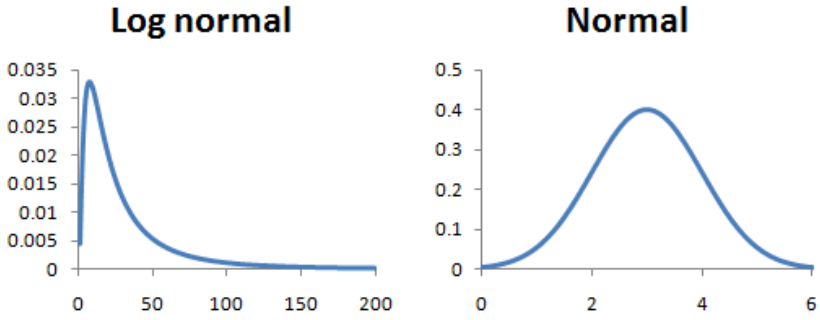
↳ Not Normal ↵

Normal data provides easy calculation.

for converting non-normal data, we apply log on original data

Case I. If transformed data is Normal then original data is log-normal

Case II. If transformed data is Not Normal then original data is not log-normal



Formula:-

PDF :-

$$F(x) = \frac{1}{x\sigma\sqrt{2\pi}} \cdot e^{\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right]}$$

where, ln = Natural logarithm

Characteristics :-

- Mean $(\mu) = e^{(\mu + \sigma^2/2)}$

Characteristics :-

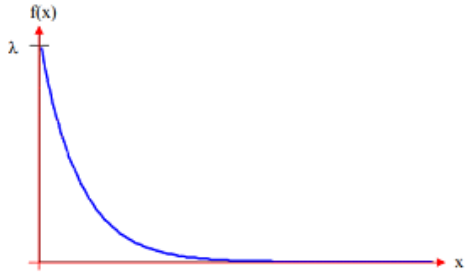
- Mean (μ) = $e^{(\mu + \sigma^2/2)}$
- Variance (σ^2) = $e^{(2\mu+\sigma^2)} \cdot e^{(\sigma^2-1)}$

Applications :-

- Finance
- Business, etc.

(4) Exponential Distribution :-

- It is used where the events occur at a constant rate.
- It is generally right skewed that starts high & decreases rapidly.



It is used to model the waiting time for a particular event to occur.

Ex- When will the next call come?
When will the next customer arrives at a store?

Formula :-

PDF :-

$$F(x) = \lambda \cdot e^{-\lambda x}, \quad x \geq 0$$

where, x - random variable

where, x = random variable
λ = rate parameter

CDF :-

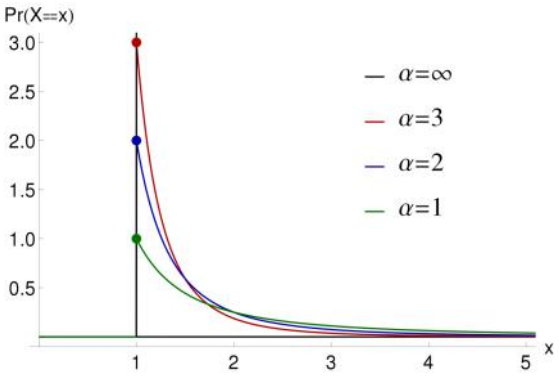
$$F(x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

Characteristics :-

- Mean (μ) = $\frac{1}{\lambda}$
- Variance (σ^2) = $\frac{1}{\lambda^2}$

⑤ Pareto distribution :-

- It has a long right tail.
- Skewness & Kurtosis depends on α parameter



Ex- Wealth of country → 80% comes from few people
20% comes from majority of people

Formula :-

PDF :-

$$F(x) = \frac{\alpha x_m^{\alpha}}{x^{\alpha+1}}$$

, $x > x_m$

α = constant

CDF :-

$$F(x) = 1 - \left[\frac{x_m}{x}\right]^{\alpha}$$